and other can be value 1 or 0. If A is 1, the output of the AND gate is 1 because both the inputs are now 1. If A is 0, the output of the AND gate is 0.Therefore, a variable ANDed with a 1 is equal to the value of the variable (A.1=A).This is easy to remember because AND operation is just like ordinary multiplication.

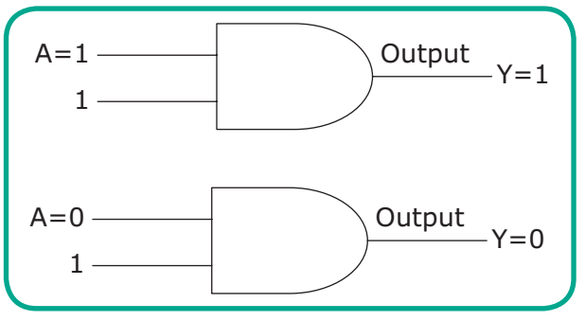


Figure 8.4 Logic gate representation of Theorem 2

Theorem 3($A+\overline{A}=0$): This theorem can be easily explained. If a variable A and its complement ($\overline{A}$) are ORed, the result is always 1. If A is 0,then $0+\overline{0} = 0+1 = 1$. If A is 1, then $1+\overline{1} = 1+0 = 1$.

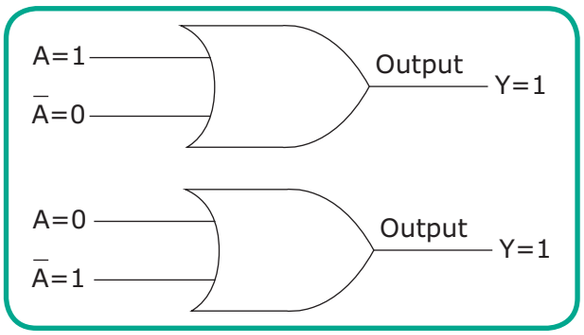


Figure 8.5 Logic gate representation of Theorem 3

Theorem 4 ($A \cdot \overline{A}=0$): This theorem states that if a variable A is ANDed with its complement, the result is zero. This is readily apparent because either A or $\overline{A}$ will always be 0. Therefore, when one of the inputs to an AND gate is 0, the output is always 0.

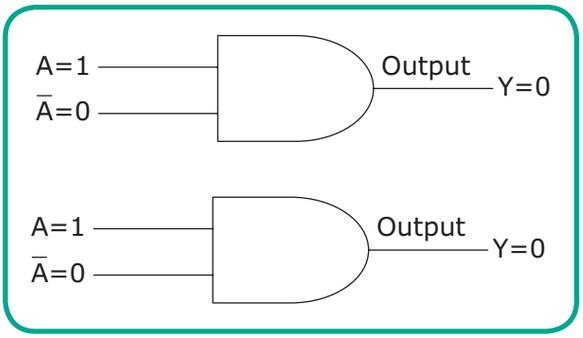


Figure 8.6 Logic gate representation of Theorem 4

Theorem 5($A+A=A$): This theorem states that when a variable A is ORed with itself, the output is equal to the variable. Thus, if A is 0, then 0+0=0, and if A is 1, then 1+1=1.



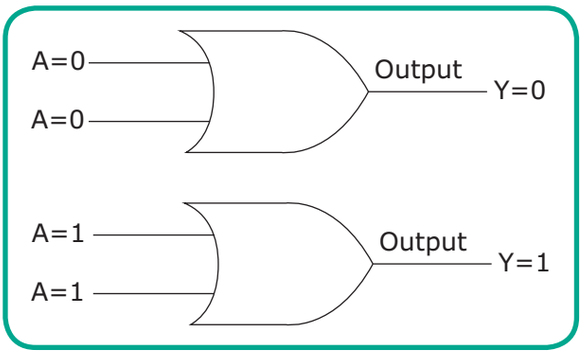


Figure 8.7 Logic gate representation of Theorem 5

Theorem 6($A.A=A$): This theorem states that when a variable A is ANDed with itself, the result is equal to the variable. For example, if A=0, then 0.0=0 and if A=1,then 1.1=1. For either case, the output of an AND gate is equal to the value of the input variable A.

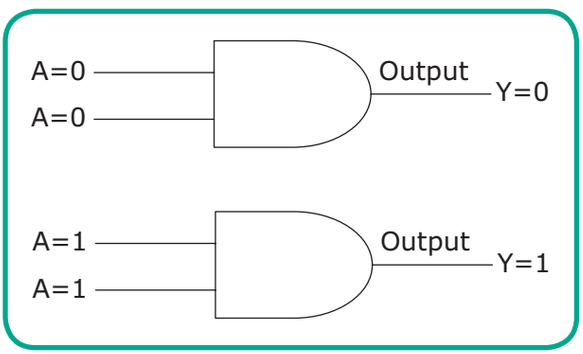


Figure 8.8 Logic gate representation of Theorem 6

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